20261119 Quiz

for (MATH102) Calculus II

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Problem 6.24. Compute

$$\iint_D f(x,y)\,\mathrm{d}A,$$

where D is the region bounded by the lines $y=1$, $x=y$, and $x=4-y$, and $f(x,y)=e^{x+y}$.

Solution. The lines $x=y$ and $x=4-y$ meet at $(2,2)$, while $y=1$ meets them at $(1,1)$ and $(3,1)$. Hence

$$D=\{(x,y)\in\mathbb{R}^2 \mid 1\le y\le 2,\ y\le x\le 4-y\}.$$

Integrating first with respect to x, we obtain

$$\begin{aligned}\iint_D e^{x+y}\,\mathrm{d}A &= \int_1^2\int_y^{4-y} e^{x+y}\,\mathrm{d}x\,\mathrm{d}y\\ &= \int_1^2 \left(e^4-e^{2y}\right)\mathrm{d}y\\ &= \left[e^4y-\frac{1}{2}e^{2y}\right]_1^2 = \boxed{\frac{e^4+e^2}{2}}.\end{aligned}$$

□

Problem 6.25. Evaluate

$$\iint_D y\,\mathrm{d}A,$$

where D is the upper half-disk $x^2+y^2\le 1$, $y\ge 0$.

Solution. For each $-1\le x\le 1$, the variable y ranges from 0 to $\sqrt{1-x^2}$. Therefore

$$\begin{aligned}\iint_D y\,\mathrm{d}A &= \int_{-1}^1\int_0^{\sqrt{1-x^2}} y\,\mathrm{d}y\,\mathrm{d}x\\ &= \frac{1}{2}\int_{-1}^1 (1-x^2)\,\mathrm{d}x\\ &= \frac{1}{2}\left[x-\frac{x^3}{3}\right]_{-1}^1 = \boxed{\frac{2}{3}}.\end{aligned}$$

□

Problem 6.26. Evaluate the integrals.

(a) $\iint_D (x^2 - y^2)\ \mathrm{d}A$, where $D = [-1, 1] \times [0, 2]$.

(b) $\iint_D x^2y^2(x + y)\ \mathrm{d}x\ \mathrm{d}y$, where $D = [0, 1] \times [0, 1]$.

Solution.

(a) By iterated integration,

$$\begin{aligned}\iint_D (x^2 - y^2)\ \mathrm{d}A &= \int_{-1}^{1}\int_{0}^{2} (x^2 - y^2)\ \mathrm{d}y\ \mathrm{d}x \\ &= \int_{-1}^{1} \left(2x^2 - \frac{8}{3}\right)\ \mathrm{d}x = \boxed{-4}.\end{aligned}$$

(b) Expanding the integrand and separating the integrals gives

$$\begin{aligned}\iint_D x^2y^2(x + y)\ \mathrm{d}x\ \mathrm{d}y &= \left(\int_0^1 x^3\ \mathrm{d}x\right)\left(\int_0^1 y^2\ \mathrm{d}y\right) \\ &\quad + \left(\int_0^1 x^2\ \mathrm{d}x\right)\left(\int_0^1 y^3\ \mathrm{d}y\right) \\ &= \frac{1}{4}\cdot\frac{1}{3} + \frac{1}{3}\cdot\frac{1}{4} = \boxed{\frac{1}{6}}.\end{aligned}$$

□

Problem 6.29. Evaluate

$$\iint_R \sin y\ \mathrm{d}A,$$

where R is the region $0 \le x \le 1$, $0 \le y \le \sqrt{x}$.

Solution. The inequality $y \le \sqrt{x}$ is equivalent to $x \ge y^2$, and $0 \le y \le 1$. Thus reversing the order of integration gives

$$\begin{aligned}\iint_R \sin y\ \mathrm{d}A &= \int_0^1 \int_{y^2}^1 \sin y\ \mathrm{d}x\ \mathrm{d}y \\ &= \int_0^1 (1 - y^2)\sin y\ \mathrm{d}y.\end{aligned}$$

Integration by parts yields

$$\int y^2 \sin y\ \mathrm{d}y = -y^2\cos y + 2y\sin y + 2\cos y.$$

Consequently,

$$\begin{aligned}\iint_R \sin y\ \mathrm{d}A &= \Big[-\cos y + y^2\cos y - 2y\sin y - 2\cos y\Big]_0^1 \\ &= \boxed{3 - 2\sin 1 - 2\cos 1}.\end{aligned}$$

□

Problem 6.30. Evaluate

$$\iint_R xe^y\ \mathrm{d}A,$$

where R is the region bounded by $y = x + 2$ and $y = x^2$.

Solution. The curves intersect where $x^2 = x + 2$, that is, at $x = -1$ and $x = 2$. Since $x^2 \le x + 2$ between these points,

$$\begin{aligned}\iint_R xe^y \, \mathrm{d}A &= \int_{-1}^{2} \int_{x^2}^{x+2} xe^y \, \mathrm{d}y \, \mathrm{d}x \\ &= \int_{-1}^{2} x\left(e^{x+2} - e^{x^2}\right) \mathrm{d}x.\end{aligned}$$

Using $\int xe^{x+2} \, \mathrm{d}x = (x-1)e^{x+2}$ and $\int xe^{x^2} \, \mathrm{d}x = \frac{1}{2}e^{x^2}$, we obtain

$$\begin{aligned}\iint_R xe^y \, \mathrm{d}A &= \left[(x-1)e^{x+2} - \frac{1}{2}e^{x^2}\right]_{-1}^{2} \\ &= \boxed{\frac{e^4 + 5e}{2}}.\end{aligned}$$

□

Problem 6.32. Consider the mapping

$$\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} u \\ v \end{pmatrix}.$$

(a) Show that the Jacobian is 1.

(b) Sketch the triangle with vertices $(1, 0)$, $(1, 1)$, and $(0, 1)$, and its image under the mapping. Show that its image has the same area.

Solution. The mapping is $\mathbf{F}(u, v) = (u + v, v)$.

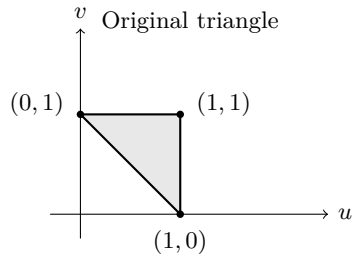
(a) Its derivative matrix is

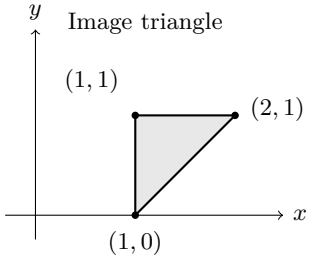
$$D\mathbf{F}(u, v) = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}, \qquad \boxed{\det D\mathbf{F}(u, v) = 1}.$$

(b) The vertices are mapped as follows:

$$(1, 0) \mapsto (1, 0), \qquad (1, 1) \mapsto (2, 1), \qquad (0, 1) \mapsto (1, 1).$$

The original triangle has horizontal and vertical legs of length 1. Its image also has a horizontal leg of length 1 and a vertical leg of length 1, as shown below.





Both triangles therefore have area

$$\boxed{\text{Area} = \frac{1}{2}}.$$

This agrees with the Jacobian $|\det D\mathbf{F}| = 1$.

□

Problem 6.34. For a mapping of the form $(x, y) = (f(u), g(v))$, verify that the Jacobian is given by

$$\det \begin{pmatrix} x_u & x_v \\ y_u & y_v \end{pmatrix} = f'(u)g'(v).$$

Solution. Since $x = f(u)$ depends only on u and $y = g(v)$ only on v,

$$x_u = f'(u), \quad x_v = 0, \quad y_u = 0, \quad y_v = g'(v).$$

Hence the derivative matrix is diagonal, and its determinant is

$$\boxed{\det \begin{pmatrix} x_u & x_v \\ y_u & y_v \end{pmatrix} = \det \begin{pmatrix} f'(u) & 0 \\ 0 & g'(v) \end{pmatrix} = f'(u)g'(v)}.$$

□

Problem 6.37. Use the change of variables $(x, y) = (r\cos\theta, r\sin\theta)$ to polar coordinates to evaluate

$$\iint_D e^{-\|\mathbf{X}\|} \, \mathrm{d}A,$$

where D is the set of points (x, y) with $a^2 \le x^2 + y^2 \le b^2$, and a, b are positive constants.

Solution. When $a \le b$, the region is the annulus $a \le r \le b$. Since $\|\mathbf{X}\| = r$ and the absolute value of the polar Jacobian is r, the change of variables formula gives

$$\begin{aligned} \iint_D e^{-\|\mathbf{X}\|} \, \mathrm{d}A &= \int_0^{2\pi} \int_a^b e^{-r} r \, \mathrm{d}r \, \mathrm{d}\theta \\ &= 2\pi \left[-(r+1)e^{-r} \right]_a^b \\ &= \boxed{2\pi \left((a+1)e^{-a} - (b+1)e^{-b} \right)}. \end{aligned}$$

If $a > b$, then D is empty and the integral is 0.

□

Problem 6.38. Use symmetries or the change of variables $(x, y) = (r\cos\theta, r\sin\theta)$ to evaluate the integrals, where D is the annulus

$$1 \le \sqrt{x^2 + y^2} \le 8.$$

(a) $\iint_D (x^2 + y^2)^p \, \mathrm{d}A$, where p is any number.

(b) $\iint_D (3 + y - x^2) \, \mathrm{d}A$.

(c) $\iint_D \ln(x^2 + y^2) \, \mathrm{d}A$.

Solution. The polar description of the annulus is $1 \le r \le 8$, $0 \le \theta \le 2\pi$, and $\mathrm{d}A = r \, \mathrm{d}r \, \mathrm{d}\theta$.

(a) For every real p,

$$\iint_D (x^2 + y^2)^p \, \mathrm{d}A = \int_0^{2\pi} \int_1^8 r^{2p+1} \, \mathrm{d}r \, \mathrm{d}\theta = 2\pi \int_1^8 r^{2p+1} \, \mathrm{d}r.$$

If $p \ne -1$, this gives

$$\boxed{\frac{\pi}{p+1} \left(8^{2p+2} - 1 \right)}.$$

If $p = -1$, the radial integrand is $1/r$, and the value is

$$\boxed{2\pi \ln 8}.$$

(b) The integral of y over D vanishes by symmetry. Moreover,

$$\begin{aligned}\iint_D 3 \, \mathrm{d}A &= 3\pi(8^2 - 1^2) = 189\pi, \\ \iint_D x^2 \, \mathrm{d}A &= \int_0^{2\pi} \cos^2\theta \, \mathrm{d}\theta \int_1^8 r^3 \, \mathrm{d}r \\ &= \pi \cdot \frac{8^4 - 1}{4} = \frac{4095\pi}{4}.\end{aligned}$$

Therefore

$$\boxed{\iint_D (3 + y - x^2) \, \mathrm{d}A = 189\pi - \frac{4095\pi}{4} = -\frac{3339\pi}{4}}.$$

(c) Since $x^2 + y^2 = r^2$, the substitution $t = r^2$ gives

$$\begin{aligned}\iint_D \ln(x^2 + y^2) \, \mathrm{d}A &= 2\pi \int_1^8 r \ln(r^2) \, \mathrm{d}r \\ &= \pi \int_1^{64} \ln t \, \mathrm{d}t \\ &= \pi \left[t \ln t - t\right]_1^{64} \\ &= \boxed{\pi \left(64 \ln 64 - 63\right)}.\end{aligned}$$

□